Transformations

Math 742

1/21/2026

In this section, we explore how transformations of random variables are used to find new distributions, both in the discrete and continuous cases.

Transformations

Suppose that X is a random variable whose PDF is $f_X(x)$ and CDF $F_X(x)$. If $Y = X^2$ or $Y = e^X$ we call Y a transformation of X. Note: These are just 2 examples of many possible transformations.

How do we find the PDF and the CDF of Y?

If X is discrete, the task is relatively simple.

X	$f_X(x)$
-2	1/4
0	1/4
2	1/2

Suppose $Y = |X|$.

Since $Y = |X|$, both $X = -2$ and $X = 2$ map to $|X| = 2$, so the total probability is the sum of the individual probabilities: $\mathbb{P}(X = -2) + \mathbb{P}(X = 2)$.

Y	$f_Y(y)$
0	1/4
2	3/4

By the same reasoning, if $Y = X^2$,

Y	$f_Y(y)$
0	1/4
4	3/4

What if X and Y are continuous?

There are 3 steps for transforming continuous random variables.

1. Let $Y = g(X)$ be a transformation of a continuous random variable X. For each $y \in \mathbb{R}$, determine the set

$$A_y = \{x \in \mathbb{R} : g(x) \leq y\}.$$

2. Compute the cumulative distribution function of Y:

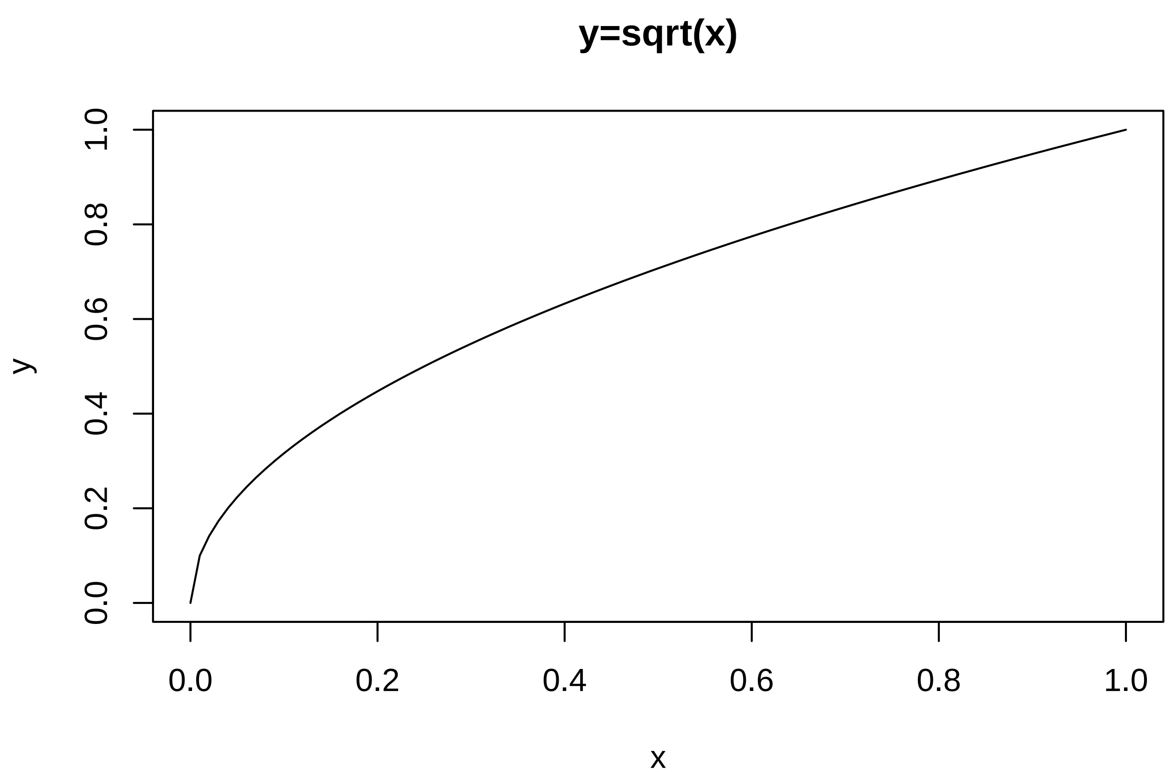
$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y) = \int_{A_y} f_X(x)\,dx.$$

3. Differentiate the CDF to obtain the probability density function of Y:

$$f_Y(y) = \frac{d}{dy} F_Y(y),$$

for all y where the derivative exists.

Example: Let X be uniform random variable with support [0,1]. This means that $f_X(x) = 1$ for $0 \leq x \leq 1$. Let $Y = \sqrt{X}$. Determine the PDF and the CDF of Y.



$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(\sqrt{X} \leq y) = \mathbb{P}(X \leq y^2) = F_X(y^2) = y^2$$

Because X is uniform on [0,1] this means $F_X(x) = x$. Thus, $F_Y(y) = y^2$.

To find the PDF we just differentiate this:

$$f_Y(y) = F'_Y(y) = 2y, 0 \leq y \leq 1$$

Inverse CDF Method:

The inverse CDF method allows us to generate random variables with a given CDF by using the inverse of the CDF. Suppose a random variable X has cumulative distribution function $F_X(x)$ and we want to generate X from a uniform random variable (i.e. $U \sim \text{Uniform}(0,1)$). In order to do this, we define

$$X = F_X^{-1}(U)$$

Applying the inverse CDF to a uniform random variable transforms U into a random variable X with the desired distribution F_X.

Proof: Assume $F_X(x)$ is continuous and strictly increasing and let $U \sim$Uniform(0,1). We want to show that,

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(F_X^{-1}(U) \leq x)$$

1. Interpretation of $F_X^{-1}(U)$:

- Let $Y = F_X^{-1}(U)$, this means that Y is defined as the inverse CDF of X applied to a uniform random variable U.
- In other words, we are transforming the uniform random variable U using the inverse CDF function F_X^{-1} to get a new random variable Y.

2. Rewrite the probability:

We are interested in the probability $\mathbb{P}(F_X^{-1}(U) \leq x)$. Since $Y = F_X^{-1}(U)$, this can be rewritten as:

$$\mathbb{P}(Y \leq x) = \mathbb{P}(F_X^{-1}(U) \leq x)$$

3. Use the definition of the inverse CDF:

- By the definition of the inverse CDF, $F_X^{-1}(U) \leq x$ is equivalent to $U \leq F_X(x)$.
- Therefore, $\mathbb{P}(F_X^{-1}(U) \leq x) = \mathbb{P}(U \leq F_X(x))$

4. Uniform distribution property:

- Since $U \sim \text{Uniform}(0,1)$, the probability that $U \leq F_X(x)$ is simply the value of $F_X(x)$, because U is uniformly distributed on (0,1).

$$\implies F_X(x) = \mathbb{P}(F_X^{-1}(U) \leq x)$$

<u>Example:</u> How would you generate a random variable X that has an exponential distribution using the inverse CDF method?

The exponential CDF is: $F_X(x) = 1 - e^{-\lambda x}, x \geq 0$. Let $u = F_X(x)$ and solve for x.

$$u = 1 - e^{-\lambda x} \implies x = \frac{-\ln(1-u)}{\lambda}$$

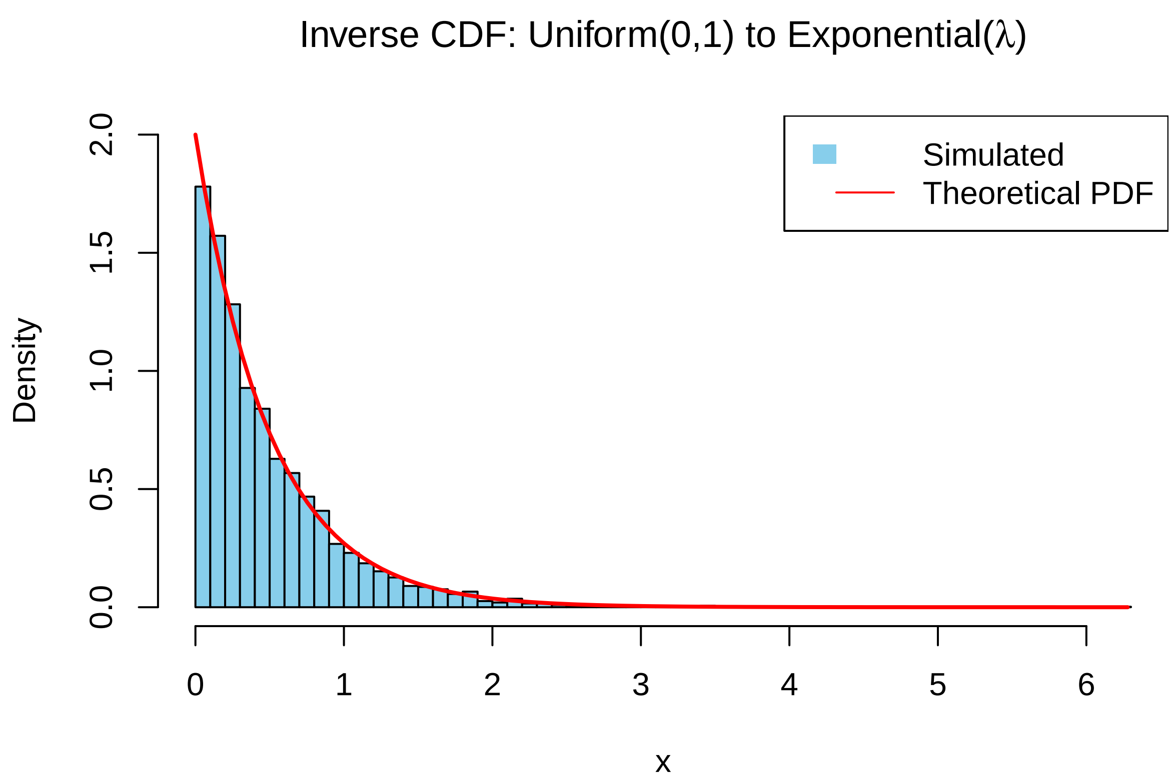
So, if $U \sim$Uniform(0,1) then,

$$X = \frac{-\ln(1-U)}{\lambda}$$

X has an exponential distribution.

So, to generate an random exponential value, we generate a uniform(0,1) random number and "plug" it in to the above formula.

```r
set.seed(742)
n <- 5000
lambda <- 2  # rate parameter (lambda)

# Step 1: Generate U ~ Uniform(0,1)
u <- runif(n, 0, 1)

# Step 2: Apply the inverse CDF of the exponential distribution
# For Exponential(lambda): F(x) = 1 - exp(-lambda x)
# Inverse: x = - (1/lambda) * log(1 - u)
x <- - (1 / lambda) * log(1 - u)

# Step 3: Compare simulated to theoretical
hist(x, breaks = 50, freq = FALSE, col = "skyblue",
     main = expression("Inverse CDF: Uniform(0,1) to Exponential(" * lambda * ")"),
     xlab = "x", ylim = c(0, lambda))
curve(lambda * exp(-lambda * x), from = 0, to = max(x), add = TRUE, lwd = 2, col = "red")
legend("topright", legend = c("Simulated", "Theoretical PDF"),
       fill = c("skyblue", NA), border = NA, lty = c(NA, 1), col = c("black", "red"))
```



Transformation Theorem (1-D): Let X be a continuous random variable with pdf $f_X(x)$ supported on an interval (a, b), where $-\infty \le a < b \le \infty$.

Let $Y = g(X)$, where $g : (a, b) \to \mathbb{R}$ is a strictly monotone (i.e. strictly increasing or decreasing), continuously differentiable function.

Define

$$\mathcal{Y} = g\big((a,b)\big).$$

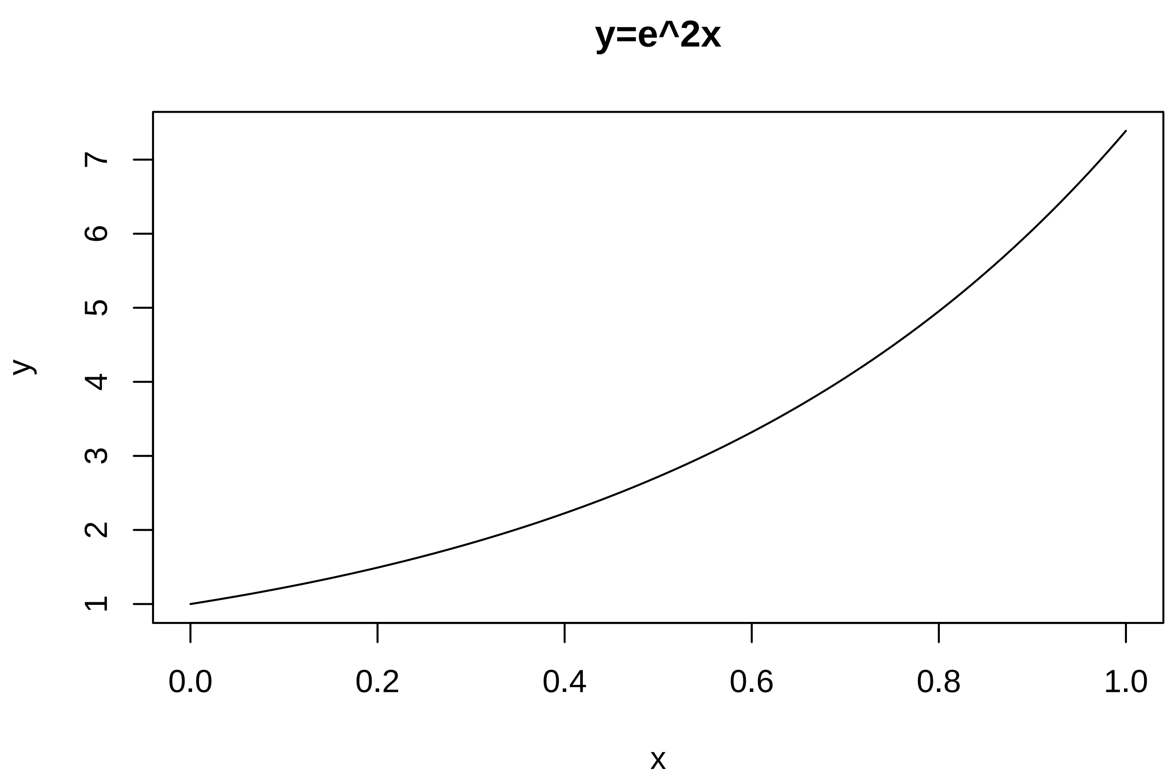
Then the inverse function g^{-1} exists on $\mathcal{Y}$ and is continuously differentiable there. The pdf of Y is given by

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right| : y \in \mathcal{Y} \\ 0 : \text{ otherwise} \end{cases}$$

This formula for $f_Y(y)$ is basically the "chain rule". f_X is the derivative of F_X and then we multiply by the derivative of the inside. Note that in order for this theorem to hold, g must be a strictly monotone function. This means that it is either strictly increasing or strictly decreasing (i.e. $u > v \implies g(u) > g(v)$ (increasing) or $u < v \implies g(u) < g(v)$ (decreasing))) [1], so we could write the PDF of Y as,

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y))\frac{d}{dy}g^{-1}(y) & y \in \mathcal{Y} \text{ if } g \text{ is increasing} \\ -f_X(g^{-1}(y))\frac{d}{dy}g^{-1}(y) & y \in \mathcal{Y} \text{ if } g \text{ is decreasing} \end{cases}$$

Example: Let $X \sim$ Uniform(0,1) and $Y = g(X) = e^{2X}$. Find the pdf $f_Y(y)$.



Find the inverse transformation, $y = e^{2x} \implies x = \frac{1}{2}\ln(y)$

So, if $x \in (0,1), y \in (e^0, e^2) = (1, e^2)$.

[1] Technically, monotone also means that it is non-decreasing, i.e. it could stay the same.

$dy/dx = 2e^{2x} = 2y$ and $dx/dy = \frac{1}{2y}$

$$f_Y(y) = f_X(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right|$$

We know that $g^{-1}(y) = \frac{1}{2}\ln(y)$ and $dx/dy = 1/2y$, so because g is increasing we don't need the absolute value vars,

$$f_Y(y) = f_X(1/2\ln(y))(1/2y)$$

Notice, that as long as x is between 0 and 1, $f_X(x) = 1$. Well, $y \in (1, e^2) \implies x$ is between 0 and 1 so as long as y is between 1 and e^2 we get,

$$f_Y(y) = \begin{cases} \frac{1}{2y} : & 1 < y < e^2 \\ 0 : & \text{otherwise} \end{cases}$$

We can verify that $f_Y(y)$ is in fact a PDF.

$$\int_1^{e^2} f_Y(y)dy = \int_1^{e^2} \frac{1}{2y}dy$$

$$= \frac{1}{2}\ln(y)\Big|_1^{e^2} = \frac{1}{2}(\ln(e^2 - \ln(1)) = \frac{1}{2}(2-0) = 1$$

Transformations in Higher Dimensions

Consider the transformation $(x, y) \mapsto (u, v)$. or equivalently $u = g_1(x, y), \qquad v = g_2(x, y)$.

Let (X, Y) have joint pdf $f_{X,Y}(x, y)$. The joint pdf of (U, V) is obtained by:

$$f_{U,V}(u, v) = f_{X,Y}(x(u, v), y(u, v))\left|\det(J)\right|,$$

where J is the **Jacobian matrix** of the inverse transformation:

$$J = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix}.$$

The absolute value of the determinant accounts for the *local area scaling* caused by the transformation.

Geometric Interpretation: The Jacobian determinant represents the scaling factor of the area (in 2D) or volume (in 3D) due to the transformation. This is why we multiply the original PDF by the absolute value of the Jacobian determinant to account for the changes in space caused by the transformation.

Example: Polar Coordinates

Suppose (X, Y) has joint pdf $f_{X,Y}(x, y)$.

Define the transformation:

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

with inverse $r = \sqrt{x^2 + y^2}, \qquad \theta = \tan^{-1}\left(\frac{y}{x}\right)$.

Compute the Jacobian:

$$J = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}.$$

The determinant is:

$$\det(J) = r(\cos^2\theta + \sin^2\theta) = r.$$

Hence,

$$f_{R,\Theta}(r,\theta) = f_{X,Y}(r\cos\theta, r\sin\theta)\, r.$$

Geometric Interpretation

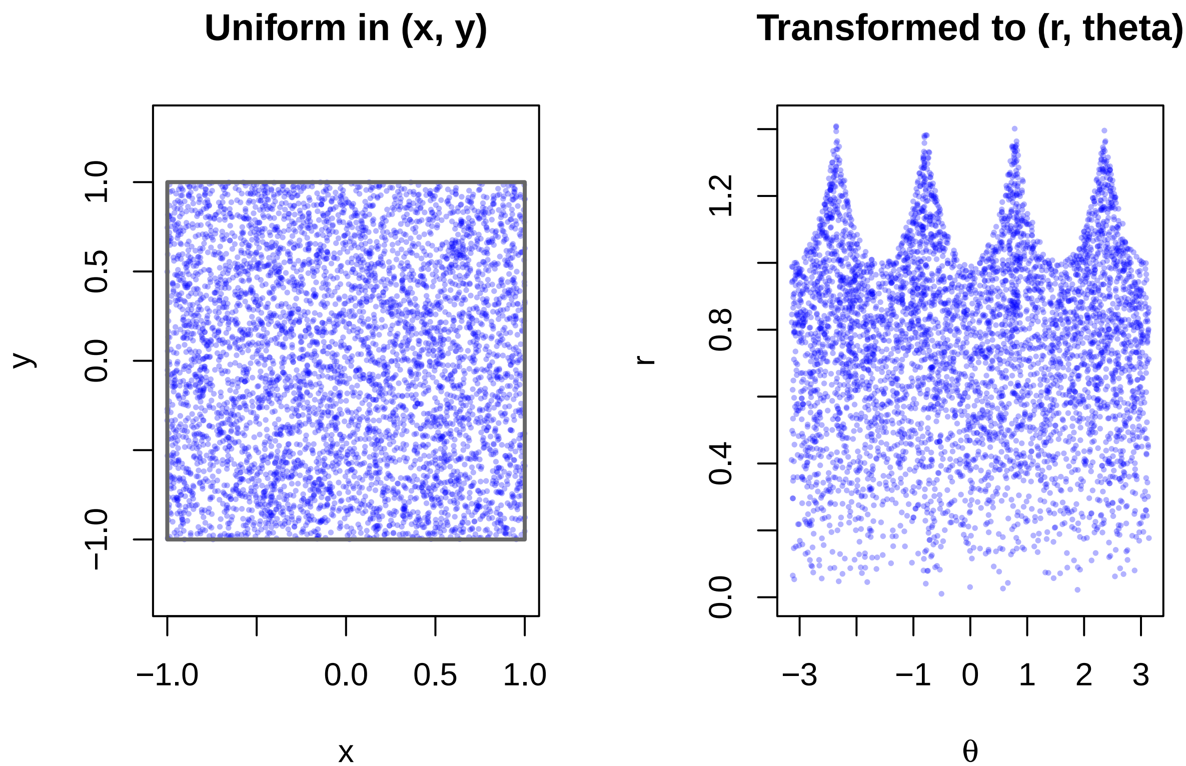
The Jacobian determinant represents how a small element of area changes under the transformation:

$$dA_{x,y} = |\det(J)|\, dA_{u,v}.$$

In the polar example, an infinitesimal rectangle in (r, θ) space has area $r\,dr\,d\theta$ in (x, y) space.

Example in R:

```r
set.seed(742)
n <- 5000
x <- runif(n, -1, 1)
y <- runif(n, -1, 1)

# Transform to polar coordinates
r <- sqrt(x^2 + y^2)
theta <- atan2(y, x)

# Set up side-by-side plots
par(mfrow = c(1, 2))

# (1) Cartesian space
plot(x, y, asp = 1, pch = 16, cex = 0.4, col = rgb(0, 0, 1, 0.3),
     main = "Uniform in (x, y)", xlab = "x", ylab = "y",
     xlim = c(-1, 1), ylim = c(-1, 1))
rect(-1, -1, 1, 1, border = "gray40", lwd = 2)

# (2) Polar coordinate space
plot(theta, r, pch = 16, cex = 0.4, col = rgb(0, 0, 1, 0.3),
     main = "Transformed to (r, theta)", xlab = expression(theta), ylab = "r",
     xlim = c(-pi, pi), ylim = c(0, sqrt(2)))
```



What is this plot showing us?

- The left panel shows points uniformly distributed in the square $[-1, 1] \times [-1, 1]$
- The right panel shows the same points after transforming to polar coordinates:

  $r = \sqrt{x^2 + y^2}$

  $\theta = \tan^{-1}(y/x)$

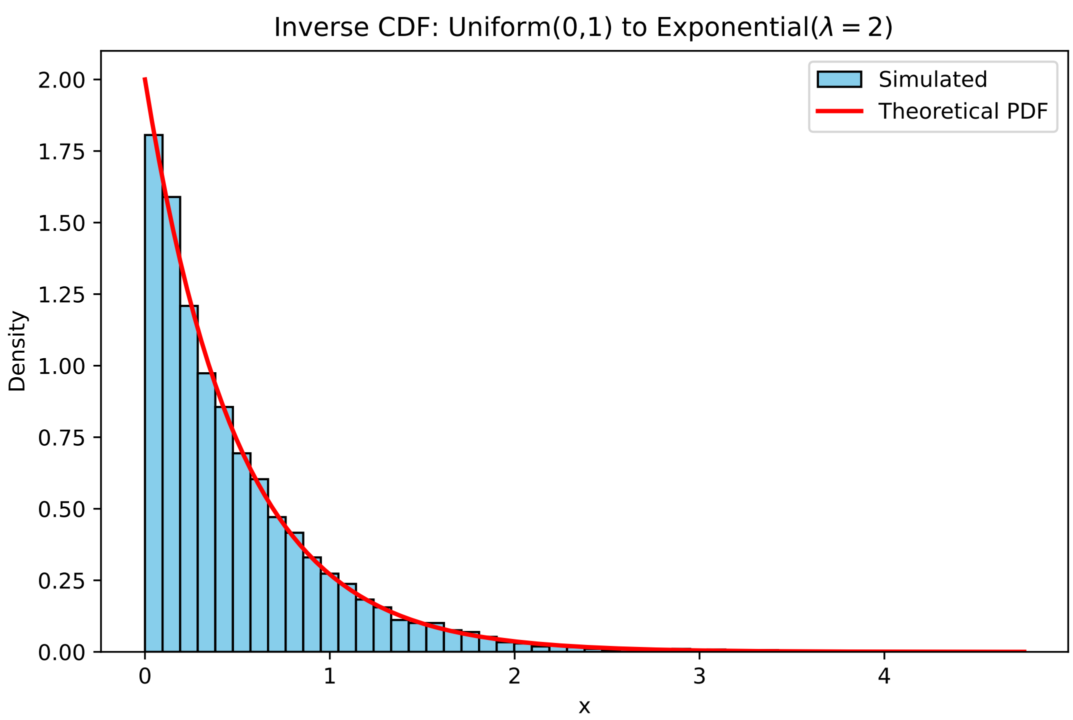
Notice that in the right panel, the points are not evenly spread over the (r, θ) plane. The **density of points increases with** r, this is because an annulus at larger r covers more area in (x, y) space. This is precisely what the **Jacobian determinant**$|\det(J)| = r$ corrects for in the PDF.

Transformation	Jacobian $\lvert\det(J)\rvert$	Interpretation
$Y = g(X)$	$\left\lvert\frac{d}{dy}g^{-1}(y)\right\rvert$	1D scaling
$(x, y) \mapsto (u, v)$	$\lvert\det J\rvert$	2D area scaling
$(x, y, z) \mapsto (u, v, w)$	$\lvert\det J\rvert$	3D volume scaling

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Step 0: Set seed and parameters
np.random.seed(742)
```

```python
n = 5000
lambda_param = 2  # rate parameter (lambda)

# Step 1: Generate U ~ Uniform(0,1)
u = np.random.uniform(0, 1, n)

# Step 2: Apply inverse CDF of Exponential(lambda)
# x = - (1 / lambda) * log(1 - u)
x = - (1 / lambda_param) * np.log(1 - u)

# Step 3: Plot histogram and theoretical density
plt.figure(figsize=(8, 5))
plt.hist(x, bins=50, density=True, color="skyblue", edgecolor="black", label="Simulated")

# Theoretical PDF
t = np.linspace(0, max(x), 500)
pdf = lambda_param * np.exp(-lambda_param * t)
plt.plot(t, pdf, color="red", linewidth=2, label="Theoretical PDF")

plt.title(r"Inverse CDF: Uniform(0,1) to Exponential($\lambda = 2$)")
plt.xlabel("x")
plt.ylabel("Density")
plt.legend()
plt.show()
```




```python
import numpy as np
import matplotlib.pyplot as plt

# Set seed and sample points
```

```python
np.random.seed(742)
n = 5000
x = np.random.uniform(-1, 1, n)
y = np.random.uniform(-1, 1, n)

# Transform to polar coordinates
r = np.sqrt(x**2 + y**2)
theta = np.arctan2(y, x)

# Set up side-by-side plots
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# (1) Cartesian space
axes[0].scatter(x, y, s=10, c='blue', alpha=0.3)
axes[0].set_title("Uniform in (x, y)")
axes[0].set_xlabel("x")
axes[0].set_ylabel("y")
axes[0].set_xlim(-1, 1)
```

```
## (-1.0, 1.0)
```

```python
axes[0].set_ylim(-1, 1)
```

```
## (-1.0, 1.0)
```

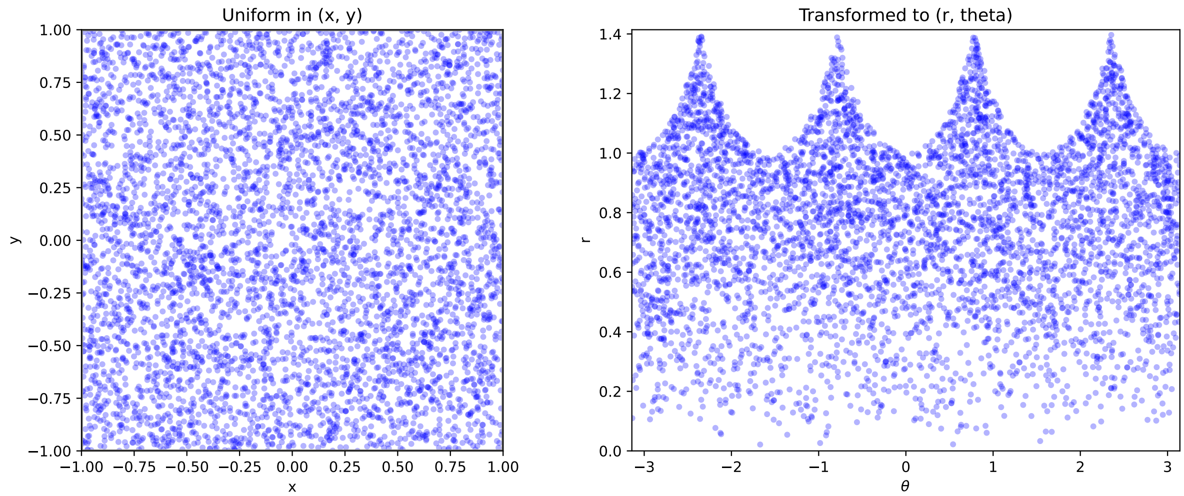
```python
axes[0].set_aspect('equal')
# Draw a rectangle around the unit square
rect = plt.Rectangle((-1, -1), 2, 2, fill=False, edgecolor='gray', linewidth=2)
axes[0].add_patch(rect)

# (2) Polar coordinate space
axes[1].scatter(theta, r, s=10, c='blue', alpha=0.3)
axes[1].set_title("Transformed to (r, theta)")
axes[1].set_xlabel(r"$\theta$")
axes[1].set_ylabel("r")
axes[1].set_xlim(-np.pi, np.pi)
```

```
## (-3.141592653589793, 3.141592653589793)
```

```python
axes[1].set_ylim(0, np.sqrt(2))
```

```
## (0.0, 1.4142135623730951)
```

```python
plt.tight_layout()
plt.show()
```


Uniform in (x, y)
y
x
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
Transformed to (r, theta)
r
θ
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−3 −2 −1 0 1 2 3